

Numerical Analysis PhD Exam, August 2019, Part 1

Do 4 (four) problems.

1. Let $A \in \mathbb{C}^{m \times n}$.

- (a) Prove or give a counterexample: $\|A\|_2 \leq \sqrt{\|A\|_\infty \|A\|_1}$. If you prove this, make sure to justify each nontrivial step.
- (b) Prove or give a counterexample: $\|A\|_2 \leq \|A\|_F$, where $\|A\|_F$ is the Frobenius norm of A . If you prove this, make sure to justify each nontrivial step.

2. This problem has two parts.

- (a) Prove that the inverse of an upper-triangular matrix is upper-triangular.
- (b) Let $A \in \mathbb{C}^{m \times n}$ with $m > n$. Consider the least-squares problem of finding $x \in \mathbb{C}^n$ that minimizes $\|Ax - b\|$ in the 2-norm. Describe a method for solving the problem efficiently, and explain (and justify) why the normal equations should not be solved.

3. Let $A \in \mathbb{C}^{m \times n}$, with $m \geq n$ and $\text{rank}(A) = p = n \geq 3$. Let $a_1, a_2, \dots$ denote the columns of A .

- (a) Using the classical Gram–Schmidt process, write out expressions for q_1, q_2, q_3 , the first three columns of Q in the QR decomposition of A .
- (b) Show the vector q_3 found in part (a) is orthogonal to both q_1 and q_2 .
- (c) Write an expression for the first Householder reflector H_1 , used to find the QR decomposition of A . Show H_1 is both unitary and Hermitian.

4. Let $A \in \mathbb{C}^{m \times m}$ be Hermitian.

- (a) Show that all eigenvalues of A are real.
- (b) Define the stationary iterative method (a.k.a. fixed point method)

$$x^{(k+1)} = Ax^{(k)} + b. \tag{1}$$

Suppose (1) has fixed-point x , namely x satisfies $x = Ax + b$. Show the iteration (1) converges to x from any starting guess $x^{(0)}$, that is $x^{(k)} \rightarrow x$ as $k \rightarrow \infty$, if and only if the eigenvalues λ_i of A satisfy $|\lambda_i| < 1$, $i = 1, \dots, m$. You may use the fact that Hermitian matrix A is unitarily diagonalizable.

5. Consider the matrix A given by

$$\begin{pmatrix} 1 & -1 & 2 & 0 \\ -1 & 4 & -1 & 1 \\ 2 & -1 & 6 & -2 \\ 0 & 1 & -2 & 4 \end{pmatrix}.$$

Suppose the eigenvalues of A are all distinct (they are) and satisfy $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$.

- (a) Show that A is positive definite.
- (b) Describe an algorithm that could be used to approximate λ_4 .
- (c) Describe algorithms that could be used to approximate λ_2, λ_3 , and their eigenvectors.